

MATH118 Homework Assignment 2

DUE

October 28, 2025, 23:55

Description

- This is an individual assignment. Please do not collaborate. Don't write anything you cannot explain.
- If you think that this document does not clearly describe the assignment, ask questions before its too late.

Instructions:

- Show all steps clearly.
- Include brief written explanations for your reasoning.
- You may discuss with classmates, but your submitted work must be your own.
- Illegible or identical-looking submissions will not be graded.

Questions

Question 1 — Discrete Random Variable

A discrete random variable X has the following probability distribution:

x	0	1	2	3	4
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

Tasks

1. Verify that this defines a valid probability distribution.
2. Compute the expected value $E[X]$ and variance $\text{Var}(X)$.
3. Find $P(X \geq 2)$.
4. Interpret the expected value if X represents the number of defects per product.

Show all steps clearly and provide brief justifications for your results.

Question 2 — Continuous Random Variable and PDF

A continuous random variable Y has pdf

$$f(y) = \begin{cases} k(1 - y^2), & -1 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Tasks

1. Find the constant k .
2. Determine the cumulative distribution function (CDF) $F(y)$.
3. Compute $P(-0.5 \leq Y \leq 0.5)$.
4. Find the mean and variance of Y .

Show all steps clearly and provide brief justifications for your results.

Question 3 — Expectation and Variance of a Function

Let X be a continuous random variable with pdf

$$f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Define $Y = X^2$.

Tasks

1. Find $E[X]$ and $\text{Var}(X)$.
2. Compute $E[Y]$ and $\text{Var}(Y)$.
3. Compare $E[Y]$ and $(E[X])^2$.

Show all steps clearly and simplify your final answers.

Submission Instructions

- This is a **handwritten assignment**.
- Use **pen or pencil** on clean, lined or plain A4 paper.
- Write your **name**, **student ID**, and **section** clearly on the **first page**.
- Show all steps, calculations, and justifications neatly.
- Maximum 1 page for each question; make sure they are numbered and ordered properly.
- After completing the work, **scan or photograph** your pages clearly and compile them into a **single PDF file**.
- The file name should follow this format: **first_name_last_name_HW2.pdf**
- No other format will be accepted.

Late Submission

- Not accepted.

Grading (Tentative)

- **Max Grade** : 100
- If you don't attend the related quiz, your homework submission will be ignored. It will be considered as 'not submitted'.